SCX (State-Conditioned eXpertise)

SCX ///

1.

1.1 Kolmogorov

1.1.1 SCX

$$R_m(s) = \mathbb{E}_{x\sim P(\cdot|s)}[\ell(f_m(x), f^*(x))]$$

Kolmogorov **Radon-Nikodym**

1.1.2 Kolmogorov $(\Omega, \mathcal{F}, P)$ X $\mathbb{E}[|X|] < \infty$ $\mathcal{G} \subseteq \mathcal{F}$ σ- P- $\mathcal{G}$- $\mathbb{E}[X|\mathcal{G}]$

$$\forall G \in \mathcal{G} : \int_G \mathbb{E}[X|\mathcal{G}]dP = \int_G XdP$$

Radon-Nikodym $\nu(G) = \int_G XdP$ $\nu \ll P|_{\mathcal{G}}$ RN $\mathbb{E}[X|\mathcal{G}]$

SCX $R_m(s)$ $\mathbb{E}[\ell(f_m(X), f^*(X))|X \in s]$ s $\mathcal{X}$ s $R_m(s) = \mathbb{E}[L_m|\mathbf{1}_s(X) = 1]$ $L_m = \ell(f_m(X), f^*(X))$

1.1.3 $P(\cdot|s)$ Regular Conditional Probability $X : \Omega \to \mathcal{X}$ σ- $\mathcal{G} \subset \mathcal{F}$ $P(\cdot|\mathcal{G})(\cdot) : \mathcal{F} \times \Omega \to [0, 1]$

1. P-a.e. $\omega P(\cdot|\mathcal{G})(\omega)$ $(\Omega, \mathcal{F})$
2. $F \in \mathcal{F} P(F|\mathcal{G})(\cdot)$ $\mathbb{E}[\mathbf{1}_F|\mathcal{G}]$

$P(\cdot|\mathcal{G})(\cdot)$

SCX s $P_{X|s}$ $\mathcal{X} = \mathbb{R}^d$ Polish Blackwell $P(x|s)$

1.1.4 ℓ $R_m(s)$ $\ell(f_m(x), f^*(x))$ $P(\cdot|s)$ (ℓ_2) (0-1) f_m f^* ℓ $f^*(x) = \arg\min_y \mathbb{E}[\ell(Y, y)|X = x]$ ()

1.2

1.2.1 Wald SCX

Abraham Wald 1950 SCX

- $\mathcal{D}$ M acquire/relabel
- $L(d, \theta)$ d θ
- $R(\pi, d) = \int_\Theta R(\theta, d)d\pi(\theta)$ π

SCX s

$$R_m(s) = \int_{\mathcal{X}} \ell(f_m(x), f^*(x))dP_{X|s}(x)$$

s

1.2.2 Berger

Berger (1985) impropriateSCX $m^*(x) = \arg\min_m \sum_s \gamma_s(x)R_m(s)$ x $\gamma_s(x)$ ””

1.2.3 PAC-Bayes $R_m(s)$ PAC-Bayes $\mathcal{Q}$ PAC-Bayes

$$\mathbb{E}_{h\sim\mathcal{Q}}[R(h)] \leq \mathbb{E}_{h\sim\mathcal{Q}}[\hat{R}(h)] + \sqrt{\frac{KL(\mathcal{Q}||\mathcal{P}) + \log(n/\delta)}{2n}}$$

SCX $w_m(x) \propto \exp(-\alpha \sum_s \gamma_s(x)\hat{R}_m(s))$ $\mathcal{Q}_x$ KL α PAC-Bayes

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1.3

1.3.1 SCX ”” s $\mathcal{X}$

$\phi : \mathcal{X} \to \mathcal{S}$

$$P(Y|X) \approx P(Y|\phi(X))$$

$\phi(X)$ Y X

1.3.2 Fisher (1922) $T(X)$ θ

$$P(X|T(X), \theta) = P(X|T(X))$$

Neyman-Fisher

$$p(x|\theta) = g(T(x), \theta) \cdot h(x)$$

SCX s $x \in s$ $R_m(s)$ x

SCX $\mathcal{S}^*$

$$R_m(s) = \mathbb{E}[\ell(f_m(X), f^*(X))|X \in s] \approx \mathbb{E}[\ell(f_m(X), f^*(X))|X = x], \quad \forall x \in s$$

1.3.3 Tishby, Pereira Bialek (1999) (IB)

$$\min_{p(\tilde{X}|X)} I(X;\tilde{X}) - \beta I(\tilde{X};Y)$$

$\tilde{X}$ X SCX $S = \phi(X)$ $\tilde{X}$ IB

$$\mathcal{L}_{IB} = I(X;S) - \beta I(S;Y)$$

- $I(X;S)$
- $I(S;Y)$
- β-

SCX IB $\mathcal{S}$ 1. $I(S;R_m)$ 2. $I(X;S)$

$\beta \to \infty$ IB x $M^* =$ $\beta \to 0$ $|S| = 1$

1.3.4 (Rate-Distortion Theory)

$$d(x,x') = |R_m(x) - R_m(x')|$$

$R_m(x) = \mathbb{E}[\ell(f_m(X), f^*(X))|X = x]$ $(\mathcal{X}, d)$ quantization

$$R(D) = \min_{P(S|X):\mathbb{E}[d(X,S)]\le D} I(X;S)$$

SCX K-means on $R(D)$

1.4 Shapley

1.4.1 SCX s

$$V(s) = \bar{r}(s) \cdot \rho(s) \cdot L(s) \cdot [1 - D(s)] \cdot \max_m SCX_m(s)$$

Shapley **Shapley** (Ghorbani & Zou, 2019)

1.4.2 Shapley Shapley (1953) $v : 2^N \to \mathbb{R}$ i

$$\phi_i(v) = \frac{1}{|N|!} \sum_{\pi \in \Pi(N)} [v(S_\pi^i \cup \{i\}) - v(S_\pi^i)]$$

S_π^i π i Shapley

1. **(Efficiency)**$\sum_{i \in N} \phi_i(v) = v(N)$
2. **(Symmetry)** $v(S \cup \{i\}) = v(S \cup \{j\})$ $S \subseteq N \setminus \{i, j\}$ $\phi_i = \phi_j$
3. **(Dummy)** $v(S \cup \{i\}) = v(S)$ S $\phi_i = 0$
4. **(Additivity)**$\phi_i(v + w) = \phi_i(v) + \phi_i(w)$

1.4.3 SCX Shapley SCX $V(s)$ Shapley Shapley $O(2^N)$ SCX

- s $O(2^{|S|})$ $|S| \ll N$
-

$V(s)$

$\bar{r}(s)$	
$\rho(s)$	
$L(s)$	
$1 - D(s)$	
$\max_m SCX_m(s)$	

Shapley

1.4.4 Data Shapley Ghorbani & Zou (2019) Shapley z_i

$$\phi_i = \frac{1}{N} \sum_{k=1}^{N} \frac{1}{\binom{N-1}{k-1}} \sum_{S \subseteq N \setminus \{i\}, |S| = k-1} [v(S \cup \{i\}) - v(S)]$$

$v(S)$ S

SCX $V(s)$ $\tilde{V}(s)$

$$V(s) \approx \mathbb{E}_{i \in s}[\phi_i] \cdot |s|$$

$V(s)$ SCX

$$v(S) \approx \prod_{s \in \mathcal{S}} g_s(|S \cap s|, \text{stats}(s))$$

v Shapley

1.5 Dawid-Skene IRT

1.5.1 Dawid-Skene Dawid & Skene (1979) J N $y_i \in \{1, \ldots, K\}$ j

$$\pi_{kl}^{(j)} = P(\ j\ k\ l)$$

$$P(|\pi, y) = \prod_{i=1}^{N} \prod_{j=1}^{J} \pi_{y_i, z_{ij}}^{(j)}$$

EM $\pi^{(j)}$ y_i

SCX $SCX_m(s) = P(\ell(f_m(x), y) < \tau | x \in s)$ Dawid-Skene

- Dawid-Skene $\pi_{kl}^{(j)}$ x
- SCX s $\pi_{kl}^{(j,s)}(x) = P(\ell(f_j(x), y) < \tau | x \in s)$

s_1, s_2 Dawid-Skene $\pi^{(j)}$ s_1 s_2

1.5.2 (IRT) Rasch (1960) IRT

$$P(Y_{ij} = 1 | \theta_i, b_j) = \frac{\exp(\theta_i - b_j)}{1 + \exp(\theta_i - b_j)}$$

θ_i i ""b_j j ""

SCX $SCX_m(s)$ Rasch

$$\log\left(\frac{SCX_m(s)}{1-SCX_m(s)}\right)=\theta_s-b_m$$

θ_s s ""b_m m

IRT 2-PL ()

$$P(Y_{ij}=1|\theta_i,a_j,b_j)=\frac{\exp(a_j(\theta_i-b_j))}{1+\exp(a_j(\theta_i-b_j))}$$

SCX a_j—— SCX $\max_m SCX_m(s)$

1.5.3 Raykar et al. (2010) SCX $SCX_m(s)$ s ——"/"

1.6 Tsybakov Huber

1.6.1 Tsybakov Tsybakov (2004) $\eta(x)=P(Y=1|X=x)$ Tsybakov $C>0$ $\alpha\geq 0$

$$P_X\left(|\eta(X)-1/2|\leq t\right)\leq Ct^{\alpha},\quad \forall t>0$$

α - $\alpha=0$ - $\alpha\to\infty$-Massart

Tsybakov $n^{-\frac{1+\alpha}{2+\alpha}}$

SCX

$$\mathrm{NoiseScore}(x_i)=r_i\cdot\frac{1}{\rho(s_i)+\varepsilon}\cdot[1-C(s_i)]$$

$C(s)$ Tsybakov α

$$C(s)\approx 1-2\cdot\mathbb{E}_{x\in s}[|\eta(x)-1/2|]$$

$C(s)\to 1$ $\alpha\to\infty$ $C(s)\to 0$ $\alpha\to 0$

s $P_X(|\eta(x) - 1/2| \leq t | x \in s) \leq C_s t^{\alpha_s}$ $C(s) = \Phi(\alpha_s)$ Φ $\Phi(\alpha) = 1 - e^{-\alpha}$ $\Phi(\alpha) = \alpha/(1+\alpha)$

1.6.2 Huber Huber (1964) ε-

$$F = (1-\varepsilon)F_0 + \varepsilon G$$

F_0 G $\varepsilon \in [0,1]$

SCX $N(s)$ s ε_s

$$P_{X|s} = (1 - N(s)) \cdot P_{X|s}^0 + N(s) \cdot P_{X|s}^{\text{noise}}$$

$P_{X|s}^0$ $P_{X|s}^{\text{noise}}$

$L(s) = C(s) \cdot [1 - N(s)]$

$$L(s) = \inf_{f \in \mathcal{F}} \left[\frac{R(f|s) - R^*(s)}{R^*(s)} \right]^{-1}$$

$R(f|s) = \mathbb{E}[\ell(f(X), Y) | X \in s] R^*(s) = \inf_f R(f|s)$ $L(s) \to 0$ $\mathcal{F}$

1.6.3 Natarajan Natarajan Natarajan (1989)

$$R(f) \leq \hat{R}_\gamma(f) + \tilde{O}\left(\sqrt{\frac{d}{n\gamma^2}}\right)$$

d Natarajan SCX $L(s)$ $\frac{d_s}{n_s}$ —— s ""

1.7 OED Bandit

1.7.1 MacKay

MacKay (1992)

$$x^* = \arg\max_x H[\theta|\mathcal{D}] - H[\theta|\mathcal{D} \cup \{x, y\}]$$

$H[\cdot]$ Shannon SCX **valuable**

SCX

$$\Delta H(s) = H[R_m|\mathcal{D}] - \mathbb{E}_{y\sim P(\cdot|s)}[H[R_m|\mathcal{D}\cup(s,y)]]$$

$V(s)$ $\Delta H(s)$

1.7.2 (OED) OED x Fisher

- **A-optimality**$\min \operatorname{tr}(I(x)^{-1})$
- **D-optimality**$\min \det(I(x)^{-1})$
- **E-optimality**$\min \lambda_{\max}(I(x)^{-1})$

SCX $V(s)$ $\bar{r}(s)\cdot L(s)$ $\bar{r}(s)$ $L(s)$ (s,y) Fisher

1.7.3 Bandit - SCX acquire **Bandit**

- (arm) s
- (reward) $Y \sim P(Y|s)$
- B

SCX $V(s)$ Upper Confidence Bound (UCB)

$$V_{\text{UCB}}(s) = \underbrace{\bar{r}(s)\cdot L(s)} + \underbrace{\rho(s)\cdot[1-D(s)]} \cdot \max_m SCX_m(s)$$

$\rho(s)$ UCB $\sqrt{\frac{\log t}{n_s}}$ $\rho(s)$ $\frac{1}{\rho(s)+\varepsilon}$

1.7.4 Thompson Sampling V(s) $\max_m SCX_m(s)$ Thompson Sampling s $m^*(s) = \arg\max_m SCX_m(s)$ ""

1.8 SCX-CompressCoreset

1.8.1 Coreset Coreset $C \subseteq \mathcal{D}$ $f \in \mathcal{F}$

$$|\sum_{i\in\mathcal{D}}\ell(f(x_i),y_i)-\sum_{j\in C}w_j\ell(f(x_j),y_j)|\leq\varepsilon\sum_{i\in\mathcal{D}}\ell(f(x_i),y_i)$$

SCX-Compress coreset

	Coreset	
Valuable	()	$w_i = 1$
Redundant	()	$w_i < 1/$
Noisy	()	$w_i = 0 \ll 1$
Expert-dependent		w_i

1.8.2 Chen et al. (2018) Coreset x_i

$$\sigma_i = \sup_{f\in\mathcal{F}}\frac{w_i|\ell(f(x_i),y_i)|}{\sum_j w_j|\ell(f(x_j),y_j)|-\varepsilon}$$

SCX $V(s)$

$$\sigma(s)\propto\frac{V(s)}{\sum_{s'}V(s')}$$

$V(s)$ coreset

1.8.3 (Wang et al. 2018) (x^*, y^*)

$$(x^*,y^*)=\arg\min_{(\tilde{x},\tilde{y})}\mathcal{L}(\theta(\tilde{x},\tilde{y}),\mathcal{D}_{\text{val}})$$

SCX-Compress Redundant

1.9 MoE

1.9.1 Jacob et al. (1991)

MoE $g(x)$ $f_m(x)$

$$f(x) = \sum_{m=1}^{M} g_m(x) \cdot f_m(x), \quad g_m(x) = \mathrm{softmax}(W_g x)_m$$

EM

SCX $m^*(x) = \arg\min_m \sum_s \gamma_s(x) R_m(s)$ Jacob hard routing vs. soft routing

1.9.2 Shazeer et al. (2017) MoE

Shazeer top-k

$$g(x) = \mathrm{softmax}(\mathrm{KeepTopK}(x \cdot W_g, k))$$

$\mathrm{KeepTopK}(v, k)_i = v_i$ v_i k $-\infty$

SCX x $\sum_s \gamma_s(x) R_m(s)$ MoE $x \to \mathrm{expert}$

1.9.3

MoE

$$\mathcal{L}_{\text{balance}} = \alpha \cdot CV(\text{expert\_load})^2$$

SCX $w_m(x) \propto \exp(-\alpha \sum_s \gamma_s(s) \hat{R}_m(s))$

1.9.4 MoE SCX

	MoE	SCX
	x	$\gamma_s(x)$
	W_g	$\hat{R}_m(s)$
	hard/top-k	
		PAC

2.

2.1

$s_1, s_2 \in \mathcal{S}$ $a, b \in \{1, \ldots, M\}$ $R_a(s_1) < R_b(s_1)$ $R_a(s_2) > R_b(s_2)$

2.1.1

1 $\preceq$ $\{1, \ldots, M\}$ $a \preceq b$ a b

2 m^* $m \neq m^*$ $s \in \mathcal{S}$ $R_{m^*}(s) \leq R_m(s)$

2.1.2

1 m^*

$s \in \mathcal{S}$

$$R_{m^*}(s) \leq R_a(s) \qquad R_{m^*}(s) \leq R_b(s)$$

s_1 $R_{m^*}(s_1) \leq R_a(s_1) < R_b(s_1)$ m^* s_1 b $\Delta_1 = R_b(s_1) - R_{m^*}(s_1) > 0$

s_2 $R_{m^*}(s_2) \leq R_b(s_2) < R_a(s_2)$ m^* s_2 a $\Delta_2 = R_a(s_2) - R_{m^*}(s_2) > 0$

$R_a(s_1) < R_b(s_1)$ $R_a(s_2) > R_b(s_2)$

$$R_a(s_1) - R_b(s_1) < 0 < R_a(s_2) - R_b(s_2)$$

$R_{m^*}(s_1) \leq R_a(s_1)$ $R_{m^*}(s_2) \leq R_b(s_2) < R_a(s_2)$

$$\begin{aligned} R_{m^*}(s_1) &\leq R_a(s_1) < R_b(s_1) \\ R_{m^*}(s_2) &\leq R_b(s_2) < R_a(s_2) \end{aligned}$$

m^*

$\mathcal{M} = \{1, \ldots, M\}$

$$\mathcal{M}^*(s) = \arg\min_{m \in \mathcal{M}} R_m(s)$$

$a \in \mathcal{M}^*(s_1)$ $b \in \mathcal{M}^*(s_2)$ $a \neq b$ $|\bigcap_{s \in \mathcal{S}} \mathcal{M}^*(s)| \leq 1$ $|\mathcal{S}| \geq 2$

$R_a(s_1) < R_b(s_1)$ $a \in \mathcal{M}^*(s_1)$ $b \notin \mathcal{M}^*(s_1)$ $R_a(s_2) > R_b(s_2)$ $b \in \mathcal{M}^*(s_2)$ $a \notin \mathcal{M}^*(s_2)$ $a \in \mathcal{M}^*(s_1)$ $b \in \mathcal{M}^*(s_2)$ $a \neq b$

$a = b$ $R_a(s_1) < R_a(s_1)$ $a \neq b$

$\mathcal{M}^*(s_1) \neq \mathcal{M}^*(s_2)\square$

2.1.3 $\mathcal{S}$ $\mathcal{S} = [0,1]$ $\exists s_1, s_2$ $R_a(s_i)$ $R_b(s_i)$ Lebesgue-

$$\mu\left(\left\{s \in \mathcal{S} : m^*(s) = \arg\min_m R_m(s)\right\}\right) < 1$$

μ $\mathcal{S}$ P_X push-forward

2.1.4 SCX - AUCF1 - -

2.2

max-residual sampling

2.2.1 $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$ $\mathcal{U}$ $x \in \mathcal{U}$

$$r(x) = \ell(f_{\text{current}}(x), \hat{y}(x))$$

$\hat{y}(x)$

max-residual $x^* = \arg\max_{x \in \mathcal{U}} r(x)$

$\tilde{r}(x)$

$$\tilde{r}(x) = r_{\text{true}}(x) + \varepsilon(x)$$

$\varepsilon(x) \sim P_{\text{noise}}(x)$

2.2.2

1. s_0 $\mathbb{E}[\varepsilon^2 | x \in s_0] \geq \sigma_0^2 > 0$
2. $B \ll |\mathcal{U}|$
3. x

2.2.3 1

π

$$J(\pi) = \mathbb{E}_{(x_1^*,\ldots,x_B^*)\sim\pi}[\sum_{t=1}^{B} \Delta R(f_t|x_t^*)]$$

$\Delta R(f_t|x) = R(f_{t-1}) - R(f_t)$ x

$r_i = r(x_i)\hat{r}_i = \tilde{r}(x_i) - \varepsilon_i$

1max-residual π_{MR} $B = 1$

$$x_{\mathrm{MR}}^* = \arg\max_{i\in\mathcal{U}} \hat{r}_i + \varepsilon_i$$

2 π^* $B = 1$

$$x_{\pi^*}^* = \arg\max_{i\in\mathcal{U}} \mathbb{E}[\Delta R(f|x_i)] = \arg\max_{i\in\mathcal{U}} \int \Delta R(f|x_i) dP_{\varepsilon_i}(\varepsilon_i)$$

Theorempair (i, j) $\hat{r}_i > \hat{r}_j$ $\mathbb{E}[\Delta R(f|x_i)] < \mathbb{E}[\Delta R(f|x_j)]$ $\pi_{\mathrm{MR}} \neq \pi^*$

1 ΔR r

$L(f(x), y)$ f_{current} Taylor

$$\Delta R(f|x) \approx r(x)^2 \cdot \left.\frac{\partial^2 \mathbb{E}[L]}{\partial f^2}\right|_{f_{\mathrm{current}}} + \sigma_y^2(x)$$

$\sigma_y^2(x)$

2 ε

$[\tilde{r}(x)]^2 = [r_{\mathrm{true}}(x) + \varepsilon]^2$

$$\mathbb{E}[[\tilde{r}(x)]^2|x] = r_{\mathrm{true}}(x)^2 + \sigma_\varepsilon^2(x)$$

$\sigma_\varepsilon^2(x)$ max-residual

3

$\mathcal{U} = \{x_1, x_2\}$ - x_1: $r_{\text{true}}(x_1) = 0.1$, $\sigma_\varepsilon^2(x_1) = 0.5$, $\tilde{r}(x_1) \approx 0.71$ - x_2: $r_{\text{true}}(x_2) = 0.5$, $\sigma_\varepsilon^2(x_2) = 0.01$, $\tilde{r}(x_2) \approx 0.50$

max-residual x_1 $\mathbb{E}[\Delta R(f|x_1)] \approx 0.01 + \sigma_y^2(x_1) \ll \mathbb{E}[\Delta R(f|x_2)] \approx 0.25 + \sigma_y^2(x_2)$

max-residual □

2.2.4 $I(X;Y|f)$ X Y max-residual $H[\hat{Y}|f,X] = \mathbb{E}[\ell(\hat{Y},Y)]$ $I(X;Y|f)$

$$I(X;Y|f) \leq H[\hat{Y}|f] - H[\hat{Y}|f,X,Y] + H[Y|f] - H[Y|f,X]$$

$H[Y|f,X]$ $I(X;Y|f)$ max-residual """"

2.3

ensemble $w_m(x)$ w_m

2.3.1 M $f_1, \ldots, f_M$

$$f_{\text{ens}}(x) = \sum_{m=1}^{M} w_m(x) f_m(x) \quad ()$$

$$\bar{f}_{\text{ens}}(x) = \sum_{m=1}^{M} \bar{w}_m f_m(x) \quad ()$$

$w_m(x) \geq 0$, $\sum_m w_m(x) = 1 \bar{w}_m \geq 0$, $\sum_m \bar{w}_m = 1$

$R(f) = \mathbb{E}[\ell(f(X), f^*(X))]$

2.3.2 $\ell(\cdot,\cdot)$ MSE, , Hinge

2.3.3 1 Jensen

1

$\bar{w} = (\bar{w}_1, \ldots, \bar{w}_M)$

$$R(\bar{f}_{\text{ens}}) = \mathbb{E}_X \left[\ell \left(\sum_m \bar{w}_m f_m(X), f^*(X) \right) \right]$$

$$R(f_{\text{ens}}) = \mathbb{E}_X \left[\ell \left(\sum_m w_m(X) f_m(X), f^*(X) \right) \right]$$

2 x

x $w^*(x) = \arg\min_{w \in \Delta^{M-1}} \ell(\sum_m w_m f_m(x), f^*(x))$

x x x

$$\ell \left(\sum_m w_m^*(x) f_m(x), f^*(x) \right) \leq \ell \left(\sum_m \bar{w}_m f_m(x), f^*(x) \right)$$

$\bar{w} \in \arg\min_{w \in \Delta^{M-1}} \ell(\sum_m w_m f_m(x), f^*(x))$

3

x

$$\mathbb{E}_X \left[\ell \left(\sum_m w_m^*(X) f_m(X), f^*(X) \right) \right] \leq \mathbb{E}_X \left[\ell \left(\sum_m \bar{w}_m f_m(X), f^*(X) \right) \right]$$

$R(f_{\text{ens}}^*) \leq R(\bar{f}_{\text{ens}})$

4SCX $w_m(x)$ $w^*(x)$

SCX $w_m(x) \propto \exp(-\alpha \sum_s \gamma_s(x) \hat{R}_m(s))$ $w^*(x)$ $\hat{R}_m(s) \xrightarrow{p} R_m(s)$ $\gamma_s(x)$ x s $w_m(x)$ $w_m^*(x)$$\square$

2.3.4

$$\Delta R = R(\bar{f}_{\text{ens}}) - R(f_{\text{ens}})$$

Taylor

$$\Delta R \geq \mathbb{E}\left[\left\langle \nabla\ell(\bar{f}_{\text{ens}}), \sum_m (\bar{w}_m - w_m(X)) f_m(X) \right\rangle\right]$$

$\delta_m(X) = \bar{w}_m - w_m(X)$

$$\Delta R \geq \sum_m \mathbb{E}[\delta_m(X) \cdot \nabla\ell(\bar{f}_{\text{ens}}) \cdot f_m(X)]$$

$\mathbb{E}[\delta_m(X)] = \bar{w}_m - \mathbb{E}[w_m(X)]$

$$\Delta R \geq \sum_m \mathbb{E}\left[\delta_m(X) \cdot \mathbb{E}[\nabla\ell(\bar{f}_{\text{ens}}) \cdot f_m(X)|X]\right]$$

$w_m(X)$ $\nabla\ell(\bar{f}_{\text{ens}}) \cdot f_m(X)$ SCX —— $\Delta R > 0$

2.3.5 $\mathcal{S} = \{\text{all}\}\gamma_s(x) \equiv 1 w_m(x) = \frac{\exp(-\alpha\hat{R}_m)}{\sum_{m'}\exp(-\alpha\hat{R}_{m'})}$ $w_m(x)$ $\Delta R = 0$

x $\mathcal{S} = \mathcal{X}\gamma_s(x) = \delta_{x,s} w_m(x) = \frac{\exp(-\alpha\hat{R}_m(x))}{\sum_{m'}\exp(-\alpha\hat{R}_{m'}(x))}\Delta R$

3.

3.1 $\phi : \mathcal{X} \to \mathbb{R}^d$

3.1.1 x $\mathbb{R}^d$

1. $\phi(x)$ $R_m(x)$
2. $\phi(x_i) \approx \phi(x_j)$ $R_m(x_i) \approx R_m(x_j)$
3. ϕ $O(d)$ $O(\log d)$

3.1.2

	d		
	$\dim(\mathcal{X})$		$\mathcal{X}$ (< 100)
PCA	(k)		
t-SNE	2-3		
UMAP			
PaCMAP		//	
			/
$[f_1(x), \ldots, f_M(x)]$	M		

3.1.3

$$\phi(x) = [\phi_{\text{feat}}(x), \phi_{\text{exp}}(x)] \in \mathbb{R}^{d_f + d_e}$$

$\phi_{\text{feat}}(x)$ $\phi_{\text{exp}}(x) = [f_1(x), \ldots, f_M(x)]$

3.2

3.2.1 PCA $\max_V \operatorname{tr}(V^\top \Sigma V)$ $\Sigma = \operatorname{Cov}(X)$

$O(nd^2 + d^3)$ $O(nd \log k)$ (SVD)

SCX PCA $R_m(x)$ PCA

$R_m(x)$ x PCA

3.2.2 t-SNE (t-distributed Stochastic Neighbor Embedding) $KL(P||Q)$ p_{ij} q_{ij} t-

$$p_{j|i} = \frac{\exp(-\|x_i - x_j\|^2 / 2\sigma_i^2)}{\sum_{k \neq i} \exp(-\|x_i - x_k\|^2 / 2\sigma_i^2)}, \quad q_{ij} = \frac{(1 + \|y_i - y_j\|^2)^{-1}}{\sum_{k \neq l} (1 + \|y_k - y_l\|^2)^{-1}}$$

$O(n^2)$ n

SCX - - d 2-3

3.2.3 UMAP (Uniform Manifold Approximation and Projection) $CE(F, G)$ F G

$O(n \log n)$

SCX - - - d 10-50

: UMAP SCX ϕ

3.2.4 PaCMAP (Pairwise Controlled Manifold Approximation) —
—

$$\text{Loss} = w_{\text{near}} \cdot \text{Loss}_{\text{near}} + w_{\text{mid}} \cdot \text{Loss}_{\text{mid}} + w_{\text{far}} \cdot \text{Loss}_{\text{far}}$$

SCX PaCMAP UMAP SCX PaCMAP

3.3

3.3.1 GMM

$$P(x) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x|\mu_k, \Sigma_k)$$

EM E-step

$$\gamma_{ik} = P(z_i = k|x_i) = \frac{\pi_k \mathcal{N}(x_i|\mu_k, \Sigma_k)}{\sum_{j=1}^{K} \pi_j \mathcal{N}(x_i|\mu_j, \Sigma_j)}$$

SCX - $\gamma_s(x) = \gamma_{ik}$ SCX $\gamma_s(x)$ - Σ_k -

- K - - $\Sigma_k = \sigma_k^2 I$

3.3.2 DBSCANDensity-Based Spatial Clustering of Applications with Noise

- ε-$N_\varepsilon(p) = \{q \in D | \text{dist}(p, q) \leq \varepsilon\}$ - $|N_\varepsilon(p)| \geq \text{MinPts}$ - -

SCX - Noise - K -

- ε MinPts - - $\gamma_s(x)$ 0 1

3.3.3 HDBSCAN (Hierarchical DBSCAN)

ε

1. MST
2. MST
3. $\lambda = 1/\text{distance}$

4.

SCX - DBSCAN - λ- -

$\mathcal{X}$ HDBSCAN GMM

3.3.4 Spectral Clustering 1. $W_{ij} = \exp(-\|x_i - x_j\|^2/2\sigma^2)$ 2. $L = D - W$ $L_{\text{sym}} = I - D^{-1/2}WD^{-1/2}$ 3. L K $U \in \mathbb{R}^{n\times K}$ 4. U K-means

SCX - - -

- $O(n^3)$ $O(n^2)$ - K - Nystroem

3.4 $\gamma_s(x)$

3.4.1 $\gamma : \mathcal{X} \times \mathcal{S} \to [0, 1]$

$$\sum_{s\in\mathcal{S}} \gamma_s(x) = 1, \quad \forall x \in \mathcal{X}$$

3.4.2 1GMM

$$\gamma_s(x) = \frac{\pi_s \mathcal{N}(\phi(x)|\mu_s, \Sigma_s)}{\sum_{t\in\mathcal{S}} \pi_t \mathcal{N}(\phi(x)|\mu_t, \Sigma_t)}$$

2Kernel

$$\gamma_s(x) = \frac{\kappa(\phi(x), \mu_s)}{\sum_{t\in\mathcal{S}} \kappa(\phi(x), \mu_t)}$$

$\kappa(x, y) = \exp(-\|x - y\|^2/2\sigma^2)\mu_s$

3t-SNE/UMAP Softmax

$$\gamma_s(x) = \mathrm{softmax}(-\beta \cdot d(\phi(x), \mu_s))_s$$

d β

4KNN

$$\gamma_s(x) = \frac{|\mathrm{KNN}(x) \cap s|}{k}$$

$\phi(x)$ k s

3.4.3 vs

	(Hard)	(Soft)

SCX $\gamma_s(x)$

$$w_m(x) \propto \exp(-\alpha \sum_s \gamma_s(x) \hat{R}_m(s))$$

3.5

3.5.1 Silhouette Score

i

$$s_i = \frac{b_i - a_i}{\max(a_i, b_i)}$$

a_i i b_i

silhouette $S(K) = \frac{1}{n} \sum_i s_i$

SCX $K^* = \arg\max_K S(K)$ silhouette $R_m(s)$

3.5.2 Gap Statistic (Tibshirani et al., 2001)

W_K $E_n^*[W_K]$

$$\mathrm{Gap}_n(K) = E_n^*[\log W_K] - \log W_K$$

K $\mathrm{Gap}(K) \geq \mathrm{Gap}(K+1) - s_{K+1}$

SCX W_K $\sum_{s \in \mathcal{S}} \mathrm{Var}_{x \in s}[R_m(x)]$

3.5.3 BIC ()

$$\mathrm{BIC}(K) = -2\log L + K \cdot \log n$$

SCX L

$$L = \prod_{i=1}^{n} \sum_{k=1}^{K} \pi_k \mathcal{N}(\phi(x_i)|\mu_k, \Sigma_k)$$

3.5.4 SCX

$$\mathrm{Var}_m(s) = \frac{1}{|s|} \sum_{x \in s} (R_m(x) - R_m(s))^2$$

$R_m(x) = \ell(f_m(x), f^*(x))$

$$H(K) = \sum_{m=1}^{M} \sum_{s \in \mathcal{S}} \mathrm{Var}_m(s)$$

$H(K+1) - H(K) < \varepsilon$ K

3.5.5 SCX

```
For K in [2, K_max]:
    1. : (X) → UMAP(X, d=min(50, n/10))
    2. : GMM(K)  HDBSCAN
    3. :
       - : Silhouette(K)
       - : H(K)
       - : BIC(K)
    4. : Score(K) = w_s  S(K) + w_h  (1-H_norm(K)) - w_b  BIC_norm(K)
 K* = argmax Score(K)
```

4.

4.1

SCX

D

1. (State Constructor)

→ → + _s(x)

{s, _s(x)}

2. (Risk Estimator)
R_m(s) = average loss of expert m on state s
SCX_m(s) = empirical reliability
: NoiseScore(x_i)

{R_m(s), SCX_m(s), NoiseScore}

3. (Value Assessor)
V(s) = r(s)(s)L(s)[1-D(s)]max_m SCX_m(s)
: Valuable / Redundant / Noisy / Expert-Dep

4. (Action Engine)
A = {acquire, relabel, downweight, discard,
route, split}
s a A

acquire/relabel downweight/discard route/split

/

```
5.  (Model Updater)
 /
 /
```

```
SCX-
(Coreset)
```

(1,)

4.2

4.2.1 →

```
:  D = {x_i, y_i, f_m(x_i)}
:
  - : state_idx[i]  {1,...,K} ()  _s(x_i) ()
  - : count[s], centroid[s], density[s]
  - : (x_i)  R^d ()
```

4.2.2 →

```
: D, {state_idx}, {f_m(x_i), y_i}
:
  - R_m(s): K × M
  - SCX_m(s): K × M
  - NoiseScore[i]: N
  - C(s): K
  - N(s): K
```

$$R_m(s) = \frac{\sum_{x_i \in s} \ell(f_m(x_i), y_i)}{|s|}$$

$$SCX_m(s) = \frac{|\{x_i \in s : \ell(f_m(x_i), y_i) < \tau\}|}{|s|}$$

$$C(s) = 1 - \frac{1}{|s|} \sum_{x_i \in s} \frac{|\ell(f_m(x_i), y_i) - \bar{\ell}(s)|}{\max \ell - \min \ell}$$

4.2.3 →

```
: {R_m(s)}, {SCX_m(s)}, {NoiseScore}, {C(s)}, {N(s)}
:
  - Value[s]: K
  - Class[s]: {'V','R','N','E'}
  - confidence[s]:


if V(s) > _v && max_m SCX_m(s) < _r:
    → Expert-dependent (E)
elif V(s) > _v && D(s) < _d:
    → Valuable (V)
elif D(s) > _d && N(s) < _n:
    → Redundant (R)
elif N(s) > _n:
    → Noisy (N)
else:
    → Expert-dependent (E) [catch-all]
```

$\tau_v, \tau_r, \tau_d, \tau_n$

4.2.4 →

```
:
  - acquire(s):  batch_size  s
  - relabel(s):  s
  - downweight(s):  s  w_s < 1
  - discard(s):  s
```

```
- route(s, m*):  s  m*
- split(s):  s
```

4.2.5 → ()

:

```
-  {f_m(x)}, {f_m(x_i)}
-  centroid
- SCX-Compress  D'
```

4.3

1. x_{new} $\phi(x_{\text{new}})$
2. $\gamma_s(x_{\text{new}})$
3. $|s|_{\text{new}} = |s|_{\text{old}} + \gamma_s(x_{\text{new}})$
4. $R_m(s)$ Welford
5. $V(s)$

- **discard**
- **split**$K = 2$
- **relabel**
- **acquire**

4.4

UMAP	$O(n \log n)$	$O(n_{\text{new}} \log n)$
GMM	$O(nKd)$	$O(Kd)$ (E-step only)
	$O(nM)$	$O(n_{\text{new}} M)$
	$O(KM)$	$O(KM)$
	$O(K)$	$O(1)$
SCX-Compress	$O(n \log n)$	$O(n_{\text{new}} \log K)$

$O(n \log n + nM)$ $O(n_{\text{new}} M)$

5. SCX

5.1

n_s s $\hat{R}_m(s)$ Hoeffding Bernstein

5.1.1 $\ell \in [0, L]$ $\delta > 0$ $1-\delta$

$$|\hat{R}_m(s) - R_m(s)| \leq L\sqrt{\frac{\log(2/\delta)}{2n_s}}$$

5.1.2 ℓ σ- MSE

$$P\left(|\hat{R}_m(s) - R_m(s)| \geq t\right) \leq 2\exp\left(-\frac{n_s t^2}{2\sigma^2}\right)$$

5.1.3 K M Bonferroni $1-\delta$

$$\max_{m,s} |\hat{R}_m(s) - R_m(s)| \leq L\sqrt{\frac{\log(2MK/\delta)}{2\min_s n_s}}$$

s n_s SCX ——

5.2

$m^*(x) = \arg\min_m \sum_s \gamma_s(x)\hat{R}_m(s)$

$$R(m^*) = \mathbb{E}[\ell(f_{m^*(X)}(X), f^*(X))]$$

5.2.1 VC $\mathcal{F}_m$ VC d_m $P_{\dim}(\mathcal{F}_m)$ $\delta > 0$ $1-\delta$

$$R(m^*) \leq \hat{R}(m^*) + \tilde{O}\left(\sqrt{\frac{\max_m d_m}{n}} + \sqrt{\frac{K\log M}{n}}\right)$$

K

5.2.2 $\mathcal{S}$ ϕ

$$R(m^*) \leq \hat{R}(m^*) + \tilde{O}\left(\sqrt{\frac{\max_m d_m + d_\phi + K \log M}{n}}\right)$$

d_ϕ Rademacher VC UMAP

5.3

5.3.1 $\mathcal{P}$

$$\inf_{\text{Alg}} \sup_{P \in \mathcal{P}} \mathbb{E}[R(\text{Alg}) - R^*] \geq \Omega\left(\sqrt{\frac{KM}{n}}\right)$$

$n/K < \log M$ SCX —— $O(\log M)$

5.3.2 -

$$\eta(\mathcal{S}) = \frac{R(\bar{f}^*_{\text{ens}}) - R(f^*_{\text{ens}})}{R(\bar{f}^*_{\text{ens}})}$$

$\eta(\mathcal{S})$

$$\eta(\mathcal{S}) \leq 1 - \frac{\min_m \sum_s P(s) R_m(s)}{\sum_s P(s) \min_m R_m(s)}$$

$\eta(\mathcal{S}) = 0$

5.4

5.4.1 NoiseScore$(x_i) = r_i \cdot \frac{1}{\rho(s_i)+\varepsilon} \cdot [1 - C(s_i)]$ Type-I/II

Tsybakov $\alpha \geq 0$

$$P(\text{false positive}) \leq C n_s^{-\frac{1+\alpha}{2+\alpha}}, \quad P(\text{false negative}) \leq C' n_s^{-\frac{1+\alpha'}{2+\alpha'}}$$

C, C' $\rho(s)$ $C(s)$

5.4.2 D(s) pairwise U-statistic

$$|\hat{D}(s) - D(s)| = O_p\left(\frac{1}{\sqrt{n_s}}\right)$$

Berry-Esseen

$$P\left(\sqrt{n_s}|\hat{D}(s) - D(s)| \geq t\right) \leq 2\exp\left(-\frac{t^2}{2\sigma_D^2}\right) + O\left(\frac{1}{\sqrt{n_s}}\right)$$

5.5 vs

5.5.1 s s_1 s_2

$$\Delta I_{\text{split}} = I(S'; R_m) - I(S; R_m)$$

S S'

$\Delta I_{\text{split}} < \log(2)/n$ 2

5.5.2 $R_{\text{gen}}(\mathcal{S})$ $\mathcal{S}$

$$R_{\text{gen}}(\mathcal{S}) \geq R^* + \exp(-I(S; R_m))$$

$R^* = \min_{m^*(x)} \mathbb{E}[\ell(f_{m^*(X)}(X), f^*(X))]$

- $R_{\text{gen}}(\mathcal{S})$ $I(S; R_m)$ - n_s

5.6

5.6.1 $\gamma_s(x)$

ℓ_∞ $\tilde{\gamma}_s(x) = \gamma_s(x) + \delta_s(x)\|\delta\|_\infty \leq \varepsilon$

$$P(m^*(x) \neq \tilde{m}^*(x)) \leq \frac{2\varepsilon}{\min_s \gamma_s(x) + \varepsilon}$$

$\gamma_s(x)$ 0

5.6.2 γ- $\gamma_s(x)$ ℓ_1

$$\tilde{\gamma}_s(x) = \frac{\gamma_s(x) + \lambda/K}{1 + \lambda}$$

$O(\varepsilon/\lambda)$

5.7 -

	K		
$K = 1$	$n_s = n$		
$K = \sqrt{n}$			-
$K = n/\log n$	$n_s = \log n$		

$n = 10^5$ $K \in [10, 100]$ $n_s \in [10^3, 10^4]$

5.8

	$O(L\sqrt{\log(MK/\delta)/n_{\min}})$	, $\ell \leq L$
	$\tilde{O}(\sqrt{(d_{\max} + K\log M)/n})$	VC $d_{\max}$, K
	$\Omega(\sqrt{KM/n})$	
	$O(n_s^{-(1+\alpha)/(2+\alpha)})$	Tsybakov $\alpha \geq 0$
	$O(\varepsilon/(\gamma_{\min} + \varepsilon))$	ℓ_∞ ε
	$\Delta I_{\text{split}} \geq \log(2)/n$	
	$R_{\text{gen}} \geq$ $R^* + \exp(-I(S; R_m))$	

$R_m(s)$		m s	1.1

$SCX_m(s)$	SCX	1.5
$V(s)$	s	1.4
$\gamma_s(x)$	x s	3.4
$\bar{r}(s)$	s	1.4
$\rho(s)$	s	1.4
$L(s)$	s	1.6
$C(s)$	s	1.6
$N(s)$	s	1.6
$D(s)$	s	5.4
$\phi(x)$		3.1
$\mathcal{S}$		1.3
M		—
$K = \|\mathcal{S}\|$		3.3
ℓ		1.1
f_m	m	—
f^*	/	1.1